PROJECT REPORT

AI-Based Water Leakage Detection and Automatic Protection System Using ESP32 with Streamlit Dashboard

Submitted By

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Github repository link: mishragaurav552006-dot/AI_Water_Leakage

Live Dashboard Link:

https://aiwaterleakage-8r7h53a3xcbbmnxcntvksc.streamlit.app/

ACKNOWLEDGEMENT

I express my sincere gratitude to the faculty members of the Department of Electronics and Communication Engineering for their valuable guidance and support throughout the development of this project. I would also like to thank my institution for providing the necessary facilities and resources required to complete this work successfully.

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ABSTRACT

Water leakage is a major issue in residential, commercial, and industrial water distribution systems, leading to water wastage and economic losses. Traditional leakage detection methods are often manual, time-consuming, and inefficient. This project proposes an **AI-Based Water Leakage Detection and Automatic Protection System** that utilizes IoT sensors, MQTT communication, and Machine Learning techniques to identify leakages in real time.

The system continuously monitors water flow, pressure, and moisture levels using sensors connected to an ESP32 microcontroller. Sensor data is transmitted through MQTT to a server where an AI model analyzes the readings and predicts leakage conditions. Upon detection of a leakage, the system automatically generates alerts, displays the leakage confidence score, activates a buzzer, switches OFF the water pump, and closes the solenoid valve to prevent water loss. A Streamlit dashboard provides real-time monitoring and historical analysis.

Keywords: Artificial Intelligence, IoT, ESP32, MQTT, Streamlit, Water Leakage Detection, Machine Learning.

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CHAPTER 1: INTRODUCTION

Water is one of the most valuable natural resources. However, a significant amount of water is wasted every year due to leakages in pipelines and water distribution systems. Conventional leakage detection systems rely heavily on manual inspection, which is inefficient and incapable of providing real-time responses.

This project presents an intelligent leakage detection system that integrates Artificial Intelligence and IoT technologies to identify, notify, and prevent water leakage automatically.

The proposed system uses:

- Flow Sensor
- Pressure Sensor
- Moisture Sensor
- ESP32 Controller
- MQTT Communication Protocol
- Machine Learning Model
- Streamlit Dashboard

The system provides continuous monitoring and automated protection actions, reducing water wastage and maintenance costs.

CHAPTER 2: LITERATURE SURVEY

Existing water leakage detection systems primarily use:

Manual Inspection

- Requires human intervention.
- High operating cost.
- Delayed response.

Sensor-Based Systems

- Detect pressure changes.

- Limited detection accuracy.
- No intelligent prediction capability.

AI-Based Systems

- Utilize historical sensor data.
- High accuracy.
- Capable of predicting leakages before severe damage occurs.

The integration of AI and IoT significantly improves reliability and response time.

CHAPTER 3: PROBLEM STATEMENT

Water leakage in pipelines often remains undetected for long periods, resulting in:

- Water wastage
- Increased utility costs
- Infrastructure damage
- Reduced system efficiency

There is a need for an intelligent system capable of automatically detecting and responding to leakage events in real time.

CHAPTER 4: OBJECTIVES

The major objectives of the project are:

1. Detect water leakages using IoT sensors.
2. Collect real-time data using ESP32.
3. Transfer sensor readings using MQTT protocol.
4. Apply AI algorithms for leakage prediction.
5. Display sensor information on a Streamlit dashboard.
6. Generate alerts upon leakage detection.
7. Automatically stop the motor.

8. Automatically close the solenoid valve.
9. Maintain leakage history and reports.

CHAPTER 5: SYSTEM ARCHITECTURE

System Components

Input Layer

- Flow Sensor
- Pressure Sensor
- Moisture Sensor

Processing Layer

- ESP32 Microcontroller
- MQTT Broker
- Machine Learning Model

Output Layer

- Streamlit Dashboard
- Relay Controlled Motor
- Solenoid Valve
- Buzzer Alarm

Data Storage Layer

- SQLite/MySQL Database

CHAPTER 6: HARDWARE REQUIREMENTS

Component	Purpose
ESP32	Main Controller

Flow Sensor	Water Flow Measurement
Pressure Sensor	Pressure Monitoring
Moisture Sensor	Leakage Detection
Relay Module	Motor Control
Solenoid Valve	Water Cut-Off
Buzzer	Alarm Generation

CHAPTER 7: SOFTWARE REQUIREMENTS

Software	Function
Python	Backend Development
Streamlit	Dashboard Development
MQTT Broker (MQTTX)	Communication
EMQX	MQTT Client
SQLite/MySQL	Data Storage
VS Code	Development Environment

CHAPTER 8: METHODOLOGY

The working methodology consists of the following steps:

Step 1

Collect sensor readings from:

- Flow Sensor
- Pressure Sensor
- Moisture Sensor

Step 2

ESP32 processes sensor values.

Step 3

Sensor data is published to MQTT broker.

Step 4

Python application subscribes to MQTT topics.

Step 5

Machine Learning model analyzes incoming data.

Step 6

Determine whether leakage exists.

Step 7

If leakage is detected:

- Display alert
- Show AI confidence score
- Turn OFF motor
- Close valve
- Activate buzzer

Step 8

Store data in database and update dashboard.

CHAPTER 9: FUNCTIONAL FLOW CHART

START
|
Initialize System
|
Read Sensors
|
Publish Data via MQTT
|
AI Analysis
|
Leakage Detected?
| Yes
v
Generate Alert
|
Display Confidence Score
|
Motor OFF
|
Valve CLOSED
|
Buzzer ON
|
Update Dashboard
|
Store Database Logs
|
END
No Leakage
|
Continue Monitoring

CHAPTER 10: IMPLEMENTATION

MQTT Message Format


```
{
  "flow_rate": 10.5,
  "pressure": 2.4,
  "moisture": 0,
  "timestamp": "2026-09-30 12:30:00"
}
```


AI Prediction Logic

Inputs:

- Flow Rate
- Pressure Value
- Moisture Level

Output:

- Leakage Probability
- Confidence Percentage
- Severity Level

Severity Classification

Level	Condition
Normal	No Leakage
Low	Minor Abnormality
Medium	Leakage Detected
High	Critical Leakage

CHAPTER 11: TEST CASES

TC ID	Test Case	Input Condition	Expected Result
TC-01	Normal Water Flow	Flow=9–11 L/min, Pressure=2.2–2.7 bar	NORMAL
TC-02	Low Flow Detection	Flow < 7 L/min	LEAKAGE
TC-03	Low Pressure Detection	Pressure < 2.0 bar	LEAKAGE
TC-04	Moisture Detection	Moisture = WET	LEAKAGE
TC-05	Multiple Abnormal Conditions	Flow + Pressure + Moisture Abnormal	High Confidence Leakage
TC-06	Recovery After Leakage	Values return to normal	NORMAL State
TC-07	AI Confidence Prediction	Leakage input data	Confidence displayed
TC-08	Motor Control	Leakage detected	Motor OFF
TC-09	Valve Control	Leakage detected	Valve CLOSED
TC-10	Buzzer Activation	NORMAL → LEAKAGE	Alarm Activated

CHAPTER 12: RESULTS AND DISCUSSION

The developed system successfully:

- Monitors water parameters continuously.
- Detects leakage conditions accurately.
- Generates instant alerts.
- Displays real-time information on dashboard.

- Automatically shuts down motor.
- Automatically closes valve.
- Reduces water wastage significantly.

Expected AI accuracy: **90%+**

Detection response time: **Below 3 seconds**

CHAPTER 13: ADVANTAGES

- Real-time monitoring
- Water conservation
- Low maintenance cost
- Automated protection mechanism
- Remote access capability
- High detection accuracy
- User-friendly dashboard
- Scalable architecture

CHAPTER 14: LIMITATIONS

- Depends on sensor accuracy.
- Requires stable Wi-Fi connection.
- Initial hardware cost.
- AI model requires sufficient training data.

CHAPTER 15: FUTURE SCOPE

Future enhancements include:

- Mobile application integration
- SMS and Email alerts

- Cloud deployment
- Smart City integration
- Advanced Deep Learning algorithms
- Predictive maintenance
- Voice assistant support

CHAPTER 16: CONCLUSION

The AI-Based Water Leakage Detection and Automatic Protection System provides an efficient and intelligent approach for monitoring water distribution networks. By combining IoT devices, MQTT communication, AI analytics, and automated protection mechanisms, the system can detect leakages in real time and take corrective actions automatically. The proposed solution helps conserve water, reduce operational costs, and improve reliability in modern water management systems.

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